Understanding Support Vector Machines: How Kernel Functions Shape Decision Boundaries

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Github: https://github.com/Abhilash2330/Machine-Learning/tree/main

1. Introduction:

Machine learning is now part of data analysis and predicting modelling. Classification algorithms have been widely used in several domains such as healthcare, finance, and malware detection. Algorithms which try to learn (provide output) for an input vector that belong to one of several categories. There are ways to classify data. Support Vector Machines are good at predicting outcomes. Can manage several types of data. They work well with data that is linear and non-linear. These machines can be used for regression and classification tasks. Support Vector Machines are versatile and widely used. They have shown to be effective, in situations.

Support vector machines create decision boundaries between the classes in a supervised machine learning algorithm. SVM works to maximize the margins between classes as opposed to other classifiers which minimize the prediction error. The margin is defined as the distance of the decision boundary to the nearest points from the classes. Maximization of the margin improves performance on unseen data.

Nevertheless, real-world data can rarely be perfectly linear. In many cases, you will find that simple straight line cannot separate the classes. To remedy this limitation, Support Vector Machines employ kernel functions. Through kernel functions, the model turns the data into higher-dimensional feature space.

Kernel functions enable SVM to capture nonlinear relationships without explicitly computing the transformation of the dataset. This technique, known as the kernel trick, makes SVM both flexible and computationally efficient for complex classification problems.

The aim of this tutorial is to investigate how different kernel functions influence the decision boundaries produced by Support Vector Machines. Using the Breast Cancer dataset from the Scikit-learn library, three commonly used kernels are examined:

- Linear Kernel
- Polynomial Kernel
- Radial Basis Function (RBF) Kernel

2. Support Vector Machines

Support Vector Machines are supervised learning algorithms designed primarily for classification tasks. The main objective of an SVM classifier is to identify an optimal hyperplane that separates data points belonging to different classes.
A hyperplane can be mathematically represented as:

$$\mathbf{w} \cdot \mathbf{x} + \mathbf{b} = \mathbf{0}$$

where:

- w represents the weight vector
- x represents the input feature vector
- b represents the bias term

In a feature space with two-dimensions, the hyperplane is a straight line separating the data into two regions. All reviewed regions have an assigned class as per predictions.

SVM chooses hyperplane that maximizes gap between two classes. The distance from the nearest observations of each class to the hyperplane is called margin.

If the classes are not linear separable, then linear SVM model will not classify them accurately. Thus, kernel functions are introduced for nonlinear classification.

3. Kernel Functions in Support Vector Machines

Kernel functions allow Support Vector Machines to perform nonlinear classification by implicitly mapping data into higher-dimensional spaces. Instead of explicitly computing new feature representations, the kernel function calculates the similarity between observations in the transformed feature space.
The kernel function can be expressed as:
$K(x_i, x_j) = \varphi(x_i) \cdot \varphi(x_j)$

Different kernel functions generate different decision boundary shapes.

→3.1 Linear Kernel: The linear kernel is the simplest kernel function used in Support Vector Machines. It assumes that the data can be separated using a straight hyperplane.
The linear kernel is defined as:
$\mathbf{K(x_i, x_j) = x_i \cdot x_j}$

However, when the dataset contains nonlinear relationships, the linear kernel may fail to capture the underlying structure of the data.

→ 3.2 Polynomial Kernel: The polynomial kernel extends the linear kernel by allowing nonlinear decision boundaries. It introduces polynomial interactions between features.
The polynomial kernel is defined as:
$\mathbf{K(x_i, x_j) = (x_i \cdot x_j + c)^d}$
where:

- c controls the influence of higher-order terms
- d represents the polynomial degree

By increasing the polynomial degree, the model can capture more complex feature interactions.

→ 3.3 Radial Basis Function Kernel: The Radial Basis Function kernel is one of the most widely used kernels in SVM classification.
The RBF kernel measures similarity between observations based on their distance in the feature space.
$\mathbf{K(x_i, x_j) = \exp(-\gamma \|x_i - x_j\|^2)}$
The parameter γ (gamma) determines how far the influence of a training sample extends. Larger gamma values create more complex decision boundaries.
The RBF kernel is highly flexible and capable of modelling complex nonlinear relationships.

4. Dataset Description:

The dataset used in this tutorial is the **Breast Cancer dataset**, which is available in the Scikit-learn machine learning library. The dataset contains measurements extracted from digitized images of breast tumour samples.

Each observation describes characteristics of cell nuclei present in the tumour images. Examples of features in the dataset include:

- mean radius.
- mean texture.
- mean perimeter.
- mean area.
- mean smoothness.

The target variable represents the diagnosis of the tumour:

- 0 – Malignant
- 1 – Benign

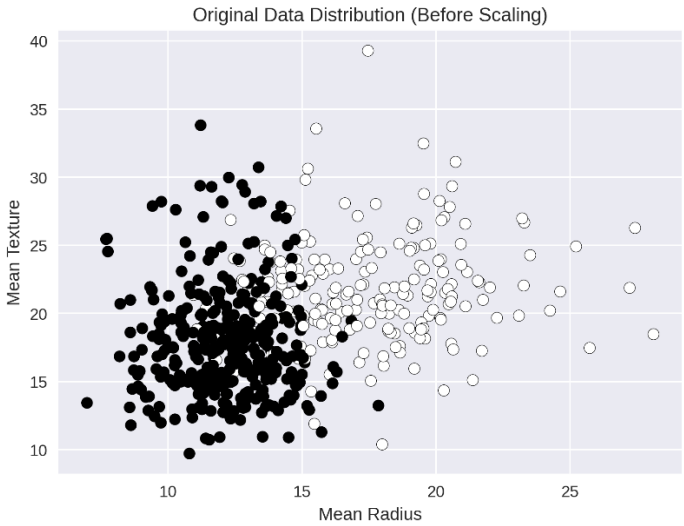
To visualize the decision boundaries clearly, only two features were selected:

- mean radius.
- mean texture.

Using two features allows the classification boundaries to be displayed in a two-dimensional feature space.

5. Data Visualization:

Before applying machine learning models, it is useful to explore the dataset visually to understand how the observations are distributed.



The visualization shows that the benign and malignant tumour samples form distinguishable clusters, although the separation between classes is not perfectly linear.

6. Data Preprocessing and Model Training:

Before training the SVM models, feature scaling was done via standardization. Support Vector Machines measure distances and are therefore sensitive to the scale of input features.

The process of standardization allows each feature to assume a mean value of zero and a standard deviation of one.

The data was scaled and split into training and testing after splitting the data by using train test split. The models were trained and evaluated using the training data and testing data, respectively.

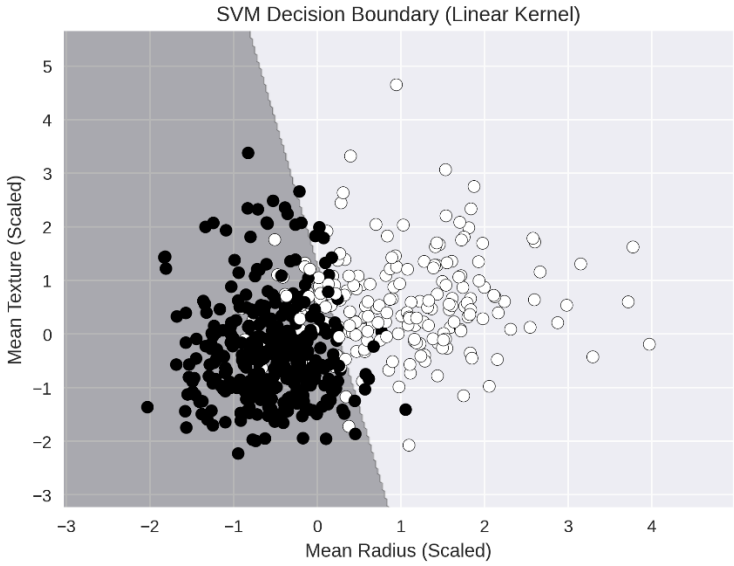
Three SVM classifiers were then trained using different kernel functions:

- Linear kernel
- Polynomial kernel
- RBF kernel

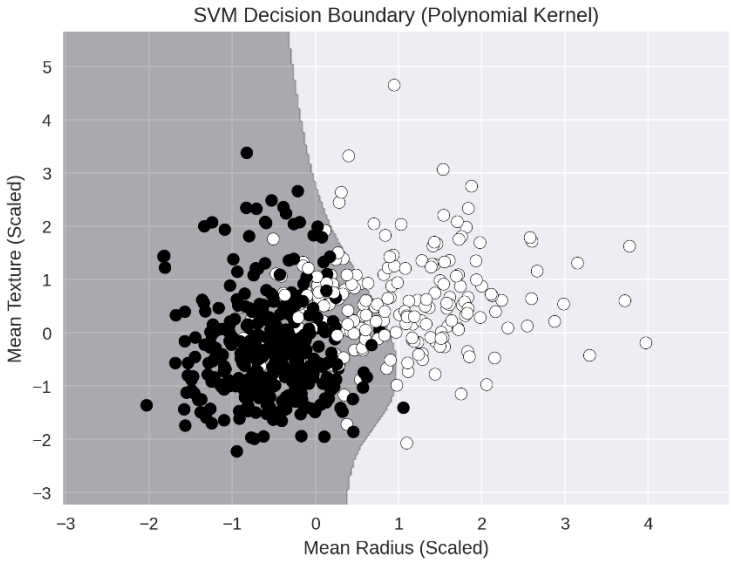
Each model was trained using the same dataset to ensure a fair comparison.

7. Decision Boundary Analysis:

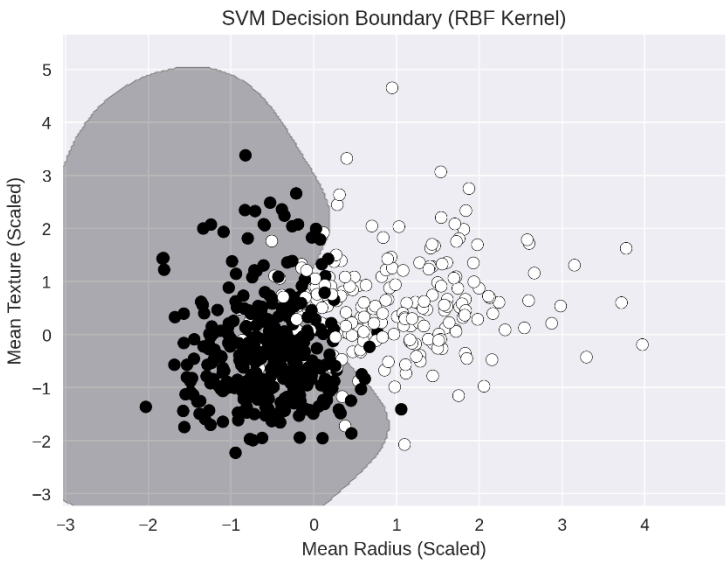
Decision boundary visualizations illustrate how different kernel functions influence classification behaviour.



The linear kernel produces a straight decision boundary separating the two classes.



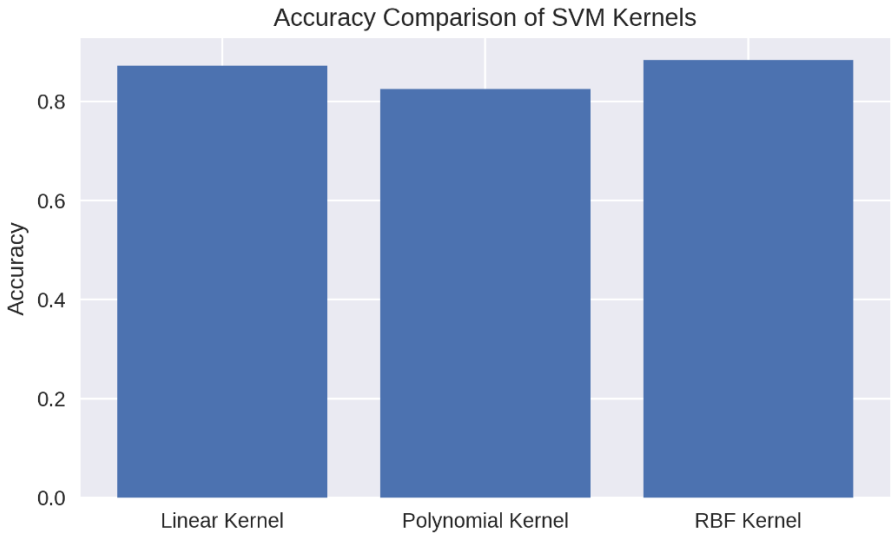
The polynomial kernel produces a curved boundary that allows the model to capture nonlinear relationships.



The RBF kernel produces the most flexible boundary, adapting to complex patterns within the dataset.

8. Model Performance Comparison:

To evaluate the performance of each model, several classification metrics were calculated including accuracy, precision, recall, and F1 score.



	Accuracy	Precision	Recall	F1 Score
Linear Kernel	0.871	0.87	0.935	0.901
Polynomial Kernel	0.825	0.781	1.0	0.877
RBF Kernel	0.883	0.892	0.925	0.908

The results indicate that nonlinear kernels often achieve slightly better performance because they capture complex feature relationships.

9. Discussion:

- The results obtained from this experiment give us a clue on the influence of kernel selection on the behaviour of Support Vector Machines.
- The linear kernel makes a straight-line boundary assuming simple linear relation between features and the target variable. Although this method is straightforward to interpret and efficient, it may not capture the structure of the data.
- The introduction of the polynomial kernel is non-linear in nature and allows the decision boundary to approximate the dataset better.
- The RBF kernel has the most flexibility as it creates localized decision region based on distance between observations.
- The performance metrics indicate that the nonlinear kernels exhibit slightly higher classification accuracy as compared to the linear kernel.
- It is important to choose a suitable kernel for applying SVMs to real-world problems as demonstrated by these results.

10. Limitations:

- Despite Robustness of SVM models, this technique has some limitation.
- Building SVM models is costly in terms of computation with large datasets.
- The SVM Models also rely heavily on using appropriate hyperparameters to perform.
- The shortcoming is SVM models compared to simpler algorithms is hard to interpret.
- Finally, we visualized use of only two variables which is a simplified version of the problem and could not achieve the complete dataset.

11. Future Work:

- Future studies could incorporate optimizing hyperparameters such as Grid Search or Cross Validation.

- Another enhancement is to train the models using all features instead of restricting the analysis to just two features.

- Comparing classification performance with other machine learning algorithms could provide further insights.

12. Conclusion:

As shown in this tutorial, the decision boundaries produced by SVMs depend on kernel functions. An observation of the effect of kernel selection on classification behaviour was made through performance comparison of linear, polynomial, and RBF kernels using the Breast Cancer dataset.

The findings reveal that nonlinear kernels enable SVM models to capture complex patterns in the dataset and improve classification performance.

It is important to understand the impact of kernel functions on decision boundaries for choosing the right model while solving a real-world machine learning problem.

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